RESULTS & VERIFICATION REPORT · VERSION 1

Adversarial Attack and Robustness Analysis of AI-Based Cataract Surgical Instrument Segmentation

DeepLabV3+ (ResNet-50) on Cataract-1K under FGSM, PGD-20 and SegPGD-20 white-box L_∞ attacks — complete experimental record, findings and verification against the problem statement

79.35%	4.35%	−95%	6 / 6
clean headline mIoU (held-out Fold 1, 561 frames)	mIoU under PGD-20 at ε = 1/255	relative fall from one intensity level per pixel	test videos: instruments degrade faster than anatomy

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Every number in this document is generated from the experiment outputs (`results/`, `runs/`) by `report/build_results_pdf.py`; re-running the script after new experiments regenerates all tables.

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1. Executive Summary

This project measured how vulnerable an AI model that segments cataract surgery video frames is to *adversarial perturbations*: small, deliberately computed changes to the input image that are invisible to a human. A DeepLabV3+ network with an ImageNet-pretrained ResNet-50 encoder was trained on the Cataract-1K segmentation dataset (3 anatomy and 9 instrument classes), evaluated on the official held-out patient fold, and then attacked with three white-box gradient attacks (FGSM, PGD-20, SegPGD-20) at perturbation budgets of 1–8 intensity levels per pixel, plus a random-noise control and several supplementary checks. In total 41 evaluation runs were made on 561 held-out frames.

79.35%	**4.35%**	**3.88%**	**5.2%**
clean mIoU (95% CI 77.34–81.01)	PGD-20, ε = 1/255	SegPGD-20, ε = 1/255	loss from random noise at ε = 8/255

Key findings

- **The clean baseline is strong.** 79.35% headline mIoU and 88.27% mean Dice on the full held-out Fold 1; every anatomy class meets or exceeds the published Cataract-1K DeepLabV3+ result. The vulnerability below cannot be blamed on a weak model.
- **The model is extremely fragile.** Changing each pixel by at most one 8-bit intensity level (ε = 1/255, PSNR ≈ 51 dB) drops headline mIoU from 79.35% to 4.35% (PGD-20) and 3.88% (SegPGD-20), a 95% relative fall. Even a quarter of one intensity level removes 73%.
- **The damage comes from the adversarial direction, not from noise.** Random ±ε noise of the same size costs at most 5.2% (ε = 8/255). FGSM costs 29–31% at every budget; iterative attacks cost 95–99.6%.
- **Instruments are more vulnerable than anatomy.** Wherever an attack is not saturated, instruments lose a larger share of their clean IoU than the cornea, pupil and lens (+3.6 to +20.7 percentage points), the bootstrap confidence intervals exclude zero, and all six test videos agree. The gap survives a control that removes hallucinated pixels.
- **The attack survives saving to an 8-bit image** (mIoU changes by at most 0.26 points after rounding).
- **Under iterative attack the model hallucinates.** At ε = 4/255 SegPGD-20 makes the model label 49.5% of all pixels as pupil (true share 6.7%) and 6.0% as lens injector (true share 0.1%).
- **More training data made the model more robust to noise but not to attacks.** Compared with an earlier run on a 908-frame subset, noise sensitivity halved while iterative-attack vulnerability was unchanged (95% fall at 1/255 in both).

Verdict on the problem statement. The question “How vulnerable is an AI-based cataract surgical instrument segmentation model to adversarial perturbations, and how significantly do such attacks affect its performance?” is answered: **extremely vulnerable**. Perturbations far below visibility remove almost all segmentation quality, the effect is directional rather than noise-like, it persists in stored 8-bit images, and surgical instruments are the most fragile part of the scene. All ten specific objectives of the Version 1 specification are met (Section 7).

2. Project Context

2.1 Problem statement

Deep-learning models for surgical video analysis classify every pixel of a frame into anatomical structures or surgical instruments. They are almost always evaluated on clean frames. Neural networks are, however, known to be susceptible to adversarial examples: inputs with carefully computed perturbations that are visually subtle but change the model's output. The central question of Version 1 is:

> **How vulnerable is an AI-based cataract surgical instrument segmentation model to adversarial perturbations, and how significantly do such attacks affect its performance?**

2.2 Research gap

The project's literature review (~65 papers; 22-paper gap matrix in `Paper_Gap_Matrix.md`) found that every neighbouring area is occupied but the intersection is empty:

- Adversarial robustness of medical segmentation is studied on radiology volumes, dental X-rays and ultrasound (e.g. Malik et al., BMVC 2024), not on intra-operative video.
- Surgical tool-segmentation robustness benchmarks are explicitly non-adversarial (SegSTRONG-C, EndoVis 2024: smoke, bleeding, low brightness).
- Adversarial work inside surgery targets detection, action triplets or robot policies on laparoscopic data (GUARDIAN, MICCAI 2024 workshop), not cataract segmentation.
- Ophthalmic adversarial studies address diagnostic imaging (fundus, OCT), never the operating-room video.
- All published Cataract-1K models, including the 2025 state of the art, are evaluated on clean frames only.

The cell "gradient-based adversarial attack × cataract surgical instrument segmentation" therefore had no matching work in the searches performed; this report fills it with a quantified vulnerability profile, including the first anatomy-versus-instrument comparison within the same surgical frame.

2.3 Research questions

Table 1. Research questions answered in Section 6.

#	Question (Version 1 specification §6)
Primary	How do adversarial attacks affect the performance and robustness of AI-based cataract surgical instrument segmentation models?
RQ1	How much does model performance decrease when adversarial perturbations are introduced?
RQ2	Which attack settings produce greater degradation?
RQ3	Are some surgical instruments or scene conditions more vulnerable than others?
RQ4	How does performance on clean inputs compare with performance on adversarial inputs?
RQ5	How does attack strength affect the trade-off between perturbation magnitude and degradation?

2.4 Scope

Table 2. Version 1 scope (specification §8 and §28). No defence was added.

In scope (Version 1)	Out of scope (reserved for Version 2)
Clean baseline training and evaluation; white-box untargeted L∞ attacks (FGSM, PGD, SegPGD); budget and iteration sweeps; per-class and anatomy-vs-instrument degradation analysis; qualitative panels	Adversarial defences, purification, adversarial training, black-box / transfer attacks, physical or patch attacks, temporal (video) attacks, clinical safety claims

3. Data

3.1 Dataset

Cataract-1K (Ghamsarian et al., *Scientific Data*, 2024) contains 1,000 cataract surgery videos recorded at Klinikum Klagenfurt. Its segmentation subset (Synapse syn53395479) provides **2,256 annotated frames from 30 videos** at 1024×768, sampled one frame every five seconds, with Supervisely polygon annotations of 3 anatomy and 9 instrument classes and five published patient-disjoint folds. Licence: CC BY 4.0.

3.2 Acquisition and integrity checks

The first web download contained images for only 908 of the 2,256 frames: Synapse packages web downloads in 1 GB parts and only the first part had been fetched. The missing 1,348 frames were downloaded directly from the Synapse folder `Annotations/Images-and-Supervisely-Annotations/<case>/img` with `download_synapse_dataset.py` (view+download access token; synapseclient verifies each file's MD5). The 30 full surgery videos in the same Synapse folder are not used. Checks performed before training:

- 2,256 images, 2,256 annotation JSONs and 2,256 rasterised masks, pairing one-to-one by file name.
- All images 1024×768 and decodable; every mask has the same size as its image.
- All 7,585 annotation objects are hole-free polygons; 0 unknown class labels.
- Annotated frames per fold reproduce the paper exactly: 561 / 459 / 420 / 385 / 431.
- Visual overlay checks of masks on newly downloaded and original frames.

The earlier 908-frame run is retained as an independent replication (Section 5.9).

3.3 Classes

Table 3. The 13-class label map (1 background + 3 anatomy + 9 instrument classes). Polygons are rasterised in z-order: anatomy first, instruments last, so an occluding instrument keeps its own label.

ID	Class	Group	Supervisely label(s)	Note
0	Background	Background	—	
1	Cornea	Anatomy	Cornea, cornea1	called “iris” in the paper
2	Pupil	Anatomy	Pupil, pupil1	
3	Lens (IOL)	Anatomy	Lens	
4	Slit/Incision Knife	Instrument	Slit Knife, Incision Knife	merged as in the paper (9 instrument classes); rare
5	Gauge	Instrument	Gauge	
6	Capsulorhexis Cystotome	Instrument	Capsulorhexis Cystotome	
7	Capsulorhexis Forceps	Instrument	Capsulorhexis Forceps	
8	Katena Forceps	Instrument	Katena Forceps	rare (spec §25)
9	Spatula	Instrument	Spatula	
10	Phacoemulsification Tip	Instrument	Phacoemulsification Tip	
11	Irrigation-Aspiration	Instrument	Irrigation-Aspiration	
12	Lens Injector	Instrument	Lens Injector	

3.4 Patient-disjoint split

Table 4. Official Cataract-1K folds. Fold 1 is the test set; one video from each other fold is used for validation (model selection), the remaining 20 videos for training. No patient appears in more than one split (asserted in code).

Fold	Videos	Annotated frames	Role
1	case5013, case5014, case5057, case5058, case5104, case5334	561	**Test** (held out)
2	case5063, case5299, case5316, **case5317**, case5319, case5335	459	Train + 1 validation video (bold)
3	case5017, case5032, case5301, case5309, **case5325**, case_53153	420	Train + 1 validation video (bold)
4	case5015, **case5300**, case5303, case5304, case5305, case5329	385	Train + 1 validation video (bold)
5	case5016, case5051, **case5072**, case5180, case5340, case5353	431	Train + 1 validation video (bold)

Table 5. Number of frames in which each class appears, per split. Every foreground class occurs in all three splits; Katena Forceps and Slit/Incision Knife are rare.

Class	Train (1,412 frames)	Val (283 frames)	Test (561 frames)
Cornea	1,412	283	561
Pupil	1,412	283	561
Lens (IOL)	362	66	107
Slit/Incision Knife	31	7	12
Gauge	271	52	103
Capsulorhexis Cystotome	53	12	20
Capsulorhexis Forceps	61	14	33
Katena Forceps	17	3	8
Spatula	412	92	212
Phacoemulsification Tip	327	70	148
Irrigation-Aspiration	280	51	120
Lens Injector	43	9	14

4. Methodology

4.1 Pipeline

Table 6. Code modules (project root).

Stage	File	What it does
Download	download_synapse_dataset.py	Fetches missing frames from Synapse (MD5-verified)
Masks	convert_supervisely_to_masks.py	Supervisely polygons → 13-class PNG label masks
Data / split	cataract_dataset.py	Official folds, patient-disjoint split, 512×512 memory-mapped cache
Model	model_builder.py	DeepLabV3+ / ResNet-50 (OS16) with ImageNet normalisation inside the model
Loss, metrics	loss.py, metrics.py	CE + 0.8·Dice; per-frame per-class TP/FP/FN; class-set means
Training	train_baseline.py	GPU augmentation, bf16 training, best-epoch selection on validation
Attacks	attacks.py	FGSM, PGD, SegPGD, random-sign noise; L∞ budget check
Experiments	run_experiments.py	41 runs on the test fold; resumable; raw per-frame counts saved
Analysis	analyze_results.py	Tables, bootstrap CIs, sign tests, per-video counts, figures
This document	report/build_results_pdf.py	Builds this PDF from the outputs above

4.2 Model

DeepLabV3+ (Chen et al., ECCV 2018) with a torchvision ResNet-50 encoder initialised from ImageNet (IMAGENET1K_V2). Output stride 16 is obtained by dilating the last residual stage. An atrous spatial pyramid pooling module (rates 6, 12, 18 plus image pooling, 256 channels) feeds a decoder that fuses the upsampled features with 48-channel projected low-level features from the first residual stage, followed by two 3×3 convolution blocks, a 1×1 classifier with 13 outputs and bilinear upsampling to 512×512 — 40.35 M parameters. ImageNet mean/std normalisation is part of the model, so the model input — and therefore the attack — lives in [0, 1] RGB pixel space. DeepLabV3+ is one of the eleven networks benchmarked in the Cataract-1K paper, and a conventional encoder–decoder avoids the known weakness of FGSM/PGD against SAM-family models, which would make a small measured degradation uninterpretable.

4.3 Training

Table 7. Baseline training configuration.

Setting	Value
Input / batch	512×512 (bilinear for frames, nearest for masks), batch 8
Loss	Cross-entropy + 0.8 × soft Dice (Cataract-1K benchmark convention); padding pixels ignored
Optimiser	AdamW, learning rate 1e-4 (encoder) / 5e-4 (decoder), weight decay 1e-4
Schedule	1-epoch linear warm-up, cosine decay per iteration; 40 epochs (as in the project plan)
Augmentation (GPU)	Horizontal (p = 0.5) and vertical (p = 0.25) flips; affine scale 0.75–1.25, rotation ±15°, shift ±10%; brightness/contrast 0.75–1.25, saturation 0.8–1.2; Gaussian blur (p = 0.2)
Precision	bf16 autocast for training; all evaluation and all attacks in fp32
Model selection	Best validation headline mIoU: epoch 36 of 40 (validation mIoU 75.82%)
Training time	54.7 min on an NVIDIA RTX 3060 Laptop GPU (6 GB)
Seed	42 (Python, NumPy, PyTorch); cuDNN benchmark mode on

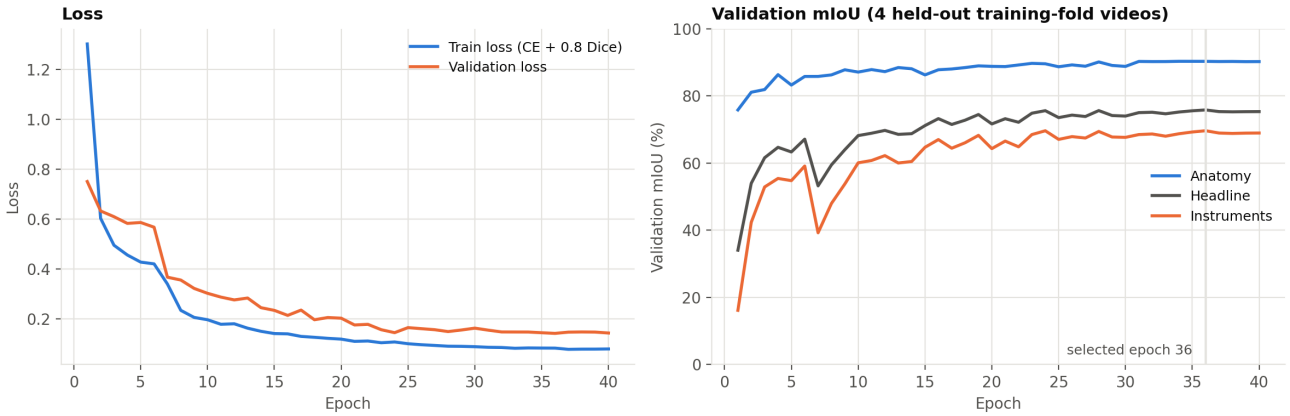


Figure 1. Training and validation loss (left) and validation mIoU per class group (right). The test fold was never used for training or model selection.

4.4 Threat model

- **White-box:** the attacker knows the architecture and weights and can compute input gradients.
- **Untargeted:** the goal is to make as many pixels as possible wrong; the attacker uses the ground-truth mask as the reference label (the standard, worst-case convention).
- **L∞-bounded in pixel space:** the perturbation δ satisfies $\|\delta\|_\infty \le \varepsilon$ and $x + \delta \in [0, 1]$, applied before ImageNet normalisation. ε is quoted in 8-bit units: ε = 1/255 means at most one intensity level per pixel and channel.
- **Digital:** perturbations are added to the decoded frame; physical, patch and temporal attacks are out of scope.

4.5 Attacks

Let $L(x) = \Sigma_{pixels}\, CE(f(x), y)$ be the summed pixel-wise cross-entropy and Π the projection onto the ε-ball intersected with [0, 1]. Summing instead of averaging does not change the sign of the gradient but keeps per-pixel gradients away from floating-point underflow.

Table 8. Attack definitions. SegPGD first pushes still-correct pixels over the decision boundary and progressively includes already-wrong pixels, so it does not waste gradient on pixels that are already misclassified, which plain pixel-averaged PGD does.

Attack	Update	Settings
FGSM (Goodfellow et al., 2015)	$x_{adv} = \Pi(x + \varepsilon \cdot \text{sign}(\nabla_x L(x)))$	one step
PGD-T (Madry et al., 2018)	$x^0 = \Pi(x + U(-\varepsilon, \varepsilon))$; $x^{t+1} = \Pi(x^t + \alpha \cdot \text{sign}(\nabla L(x^t)))$	T = 20 (ablation 5–40), α = ε/4, random start
SegPGD-T (Gu et al., 2022)	same, with $L_t = (1 - \lambda_t)\, \Sigma_{correct}\, CE + \lambda_t\, \Sigma_{wrong}\, CE$, $\lambda_t = (t - 1)/(2T)$	T = 20 (ablation 5–40), α = ε/4, random start
Random noise (control)	$x_{adv} = \Pi(x + \varepsilon \cdot s)$, $s \in \{-1, +1\}$ uniformly at random	no gradient information

Every attacked batch is checked at run time: the maximum measured $\|\delta\|_\infty$ must not exceed ε and every pixel must stay in [0, 1]. In all 40 attacked runs the measured maximum equals ε exactly (Appendix D).

4.6 Experiment grid

Table 9. The 41 evaluation runs.

Group	Runs	Budgets ε (/255)	Purpose
Clean	1	—	Reference performance
Main grid (specification)	12	1, 2, 4, 8 × {FGSM, PGD-20, SegPGD-20}	Core measurement
Noise control	4	1, 2, 4, 8	Is the damage adversarial or just noise?
Step ablation (specification)	6	4 (T = 5, 10, 40; T = 20 reused)	Effect of iterations
Step ablation (supplementary)	6	1 (T = 5, 10, 40)	Same, before saturation
Sub-8-bit budgets (supplementary)	6	¼, ½ × {FGSM, PGD-20, SegPGD-20}	Where does the collapse start?
8-bit quantised (supplementary)	6	1, 2 × {FGSM, PGD-20, SegPGD-20}	Does it survive saving as PNG?
Total	**41**		561 test frames each

4.7 Metrics and statistics

- **Per-class IoU and Dice** from pixel counts pooled over all test frames: IoU = TP/(TP+FP+FN), Dice = 2TP/(2TP+FP+FN).
- **Headline mIoU / mean Dice:** mean over the 10 non-rare foreground classes present in the test ground truth. Rare classes (Slit/Incision Knife, Katena Forceps) are reported separately (spec §25). Because ground truth does not change under attack, clean and attacked runs average over exactly the same classes.
- **Anatomy / instrument mIoU:** the same, restricted to the 3 anatomy or the 7 non-rare instrument classes.
- **Degradation:** $\Delta\text{mIoU} = \text{mIoU}_{\text{clean}} - \text{mIoU}_{\text{adv}}$; relative fall = $\Delta\text{mIoU} / \text{mIoU}_{\text{clean}} \times 100$; robustness ratio = $\text{mIoU}_{\text{adv}} / \text{mIoU}_{\text{clean}}$.
- **Correct→wrong:** share of pixels the clean model classified correctly that the attack made wrong.
- **PSNR** of the adversarial frame relative to the clean frame (perceptibility).
- **Confidence intervals:** paired frame bootstrap, 2,000 resamples, the same resampled frames for clean and attacked runs, so CIs on differences are paired.
- **Consistency:** per-frame one-sided sign test on robustness ratios, and the number of the 6 test videos in which the effect has the same direction (the conservative evidence, as frames within a video are correlated).
- **Object size:** Spearman ρ across classes between mean ground-truth area per frame and relative fall.

4.8 Corrections made to the pre-existing code

Table 10. Corrections to the code base made before the reported experiments.

Issue found during the audit	Consequence if left	Fix
test_fold=1 indexed a 0-based dict, so the test set was the paper's Fold 2 (459 frames)	Not comparable to the benchmark; spec violated	Folds renumbered 1–5; sizes asserted against the paper
model_builder.py built torchvision DeepLabV3, not DeepLabV3+	Specified architecture not implemented	DeepLabV3+ built from ResNet-50 + ASPP + low-level decoder
ImageNet normalisation applied in the dataset	ε would be defined in normalised space	Normalisation moved inside the model
Dice loss clamped labels, mapping the ignore label 255 to class 12	Padding trained as Lens Injector	Ignore pixels masked from both loss terms (unit-tested)
Validation used a whole fold (6 videos)	Spec requires 20 train / 4 val videos	One validation video per non-test fold
mIoU over every class with any prediction, incl. background and absent classes	Class set would change under attack	Class set fixed from test ground truth; rare classes separate
Whole dataset held in RAM	Swapping on a 16 GB laptop	Memory-mapped 512×512 cache, verified pixel-identical

5. Results

5.1 Clean baseline

Table 11. Clean performance on the held-out Fold 1 (561 frames, 6 patients).

Metric	Value
Headline mIoU (95% CI)	79.35 (77.34–81.01)
Headline mean Dice	88.27
Anatomy mIoU / Dice	87.84 / 93.49
Instrument mIoU / Dice (non-rare)	75.71 / 86.03
Instrument mIoU incl. rare classes	68.14
mIoU over all 12 foreground classes	73.06
Pixel accuracy	98.50

Table 12. Clean per-class results. Katena Forceps is largely missed even without attack (8 test frames); see Figure 8, row 3.

ID	Class	Group	Test frames	IoU %	Dice %
0	Background	Background	561	98.92	99.46
1	Cornea	Anatomy	561	84.55	91.63
2	Pupil	Anatomy	561	92.42	96.06
3	Lens (IOL)	Anatomy	107	86.55	92.79
4	Slit/Incision Knife *(rare)*	Instrument	12	79.16	88.37
5	Gauge	Instrument	103	70.37	82.61
6	Capsulorhexis Cystotome	Instrument	20	64.64	78.53
7	Capsulorhexis Forceps	Instrument	33	74.96	85.69
8	Katena Forceps *(rare)*	Instrument	8	4.08	7.85
9	Spatula	Instrument	212	74.36	85.30
10	Phacoemulsification Tip	Instrument	148	82.93	90.67
11	Irrigation-Aspiration	Instrument	120	84.12	91.37
12	Lens Injector	Instrument	14	78.61	88.03

Table 13. Comparison with the published benchmark: Cataract-1K paper (Sci. Data 2024), Table 7, DeepLabV3+ with ResNet-50, anatomy-plus-instrument task, where all instruments form one class (ours merged the same way).

Structure	Paper IoU	Ours IoU	Paper Dice	Ours Dice
Iris (dataset label: Cornea)	79.97	84.55	88.83	91.63
Pupil	86.24	92.42	92.61	96.06
Lens (IOL)	77.23	86.55	87.12	92.79
Instruments (merged)	53.53	79.34	69.67	88.48

The baseline meets or exceeds the published DeepLabV3+ on every structure. The comparison is indicative, not like-for-like: the paper reports a 5-fold cross-validation average with a different training recipe, whereas this project evaluates the single official Fold 1. The large merged-instrument gap is therefore *not* claimed as an improvement over the benchmark; the conclusion drawn is only that the baseline is well trained, so the vulnerability measured below is not an artefact of a weak model.

5.2 Main grid: clean versus attacked

Table 14. Main grid on the test fold. PGD/SegPGD: 20 steps, α = ε/4, random start. 95% CIs from a paired frame bootstrap. Anatomy / Instr. = group mIoU %.

Attack	ε	mIoU % (95% CI)	ΔmIoU pp	Rel. fall %	Robust. ratio	mDice %	Anat. mIoU	Instr. mIoU	Correct→ wrong %	PSNR dB
Clean	—	79.35 (77.34–81.01)	—	—	1.000	88.27	87.84	75.71	—	—
Noise (control)	1/255	79.17 (77.19–80.84)	0.18	0.2	0.998	88.16	87.78	75.48	0.1	48.2
Noise (control)	2/255	78.93 (76.92–80.61)	0.43	0.5	0.995	87.99	87.53	75.24	0.2	42.2
Noise (control)	4/255	78.15 (76.17–79.72)	1.20	1.5	0.985	87.50	86.98	74.36	0.4	36.2
Noise (control)	8/255	75.20 (73.21–76.82)	4.15	5.2	0.948	85.58	85.38	70.84	0.7	30.2
FGSM	1/255	56.28 (54.00–58.29)	23.07	29.1	0.709	71.45	69.24	50.73	2.7	48.2
FGSM	2/255	55.49 (53.24–57.50)	23.86	30.1	0.699	70.77	69.09	49.66	2.7	42.2
FGSM	4/255	55.81 (53.54–57.87)	23.54	29.7	0.703	71.03	69.92	49.76	2.6	36.2
FGSM	8/255	55.06 (52.89–57.07)	24.29	30.6	0.694	70.27	70.73	48.35	2.6	30.2
PGD-20	1/255	4.35 (3.71–4.98)	75.00	94.5	0.055	8.04	9.73	2.04	62.5	51.2
PGD-20	2/255	1.57 (1.25–1.92)	77.78	98.0	0.020	3.04	3.62	0.69	70.5	45.6
PGD-20	4/255	0.84 (0.60–1.11)	78.51	98.9	0.011	1.66	1.84	0.41	69.1	39.8
PGD-20	8/255	0.39 (0.24–0.62)	78.96	99.5	0.005	0.77	0.83	0.20	65.5	33.8
SegPGD-20	1/255	3.88 (3.39–4.42)	75.47	95.1	0.049	7.28	7.71	2.24	68.2	51.2
SegPGD-20	2/255	1.38 (1.10–1.68)	77.97	98.3	0.017	2.70	2.75	0.79	76.5	45.6
SegPGD-20	4/255	0.68 (0.49–0.93)	78.67	99.1	0.009	1.34	1.41	0.36	75.7	39.7
SegPGD-20	8/255	0.34 (0.23–0.52)	79.01	99.6	0.004	0.68	0.71	0.18	74.0	33.8

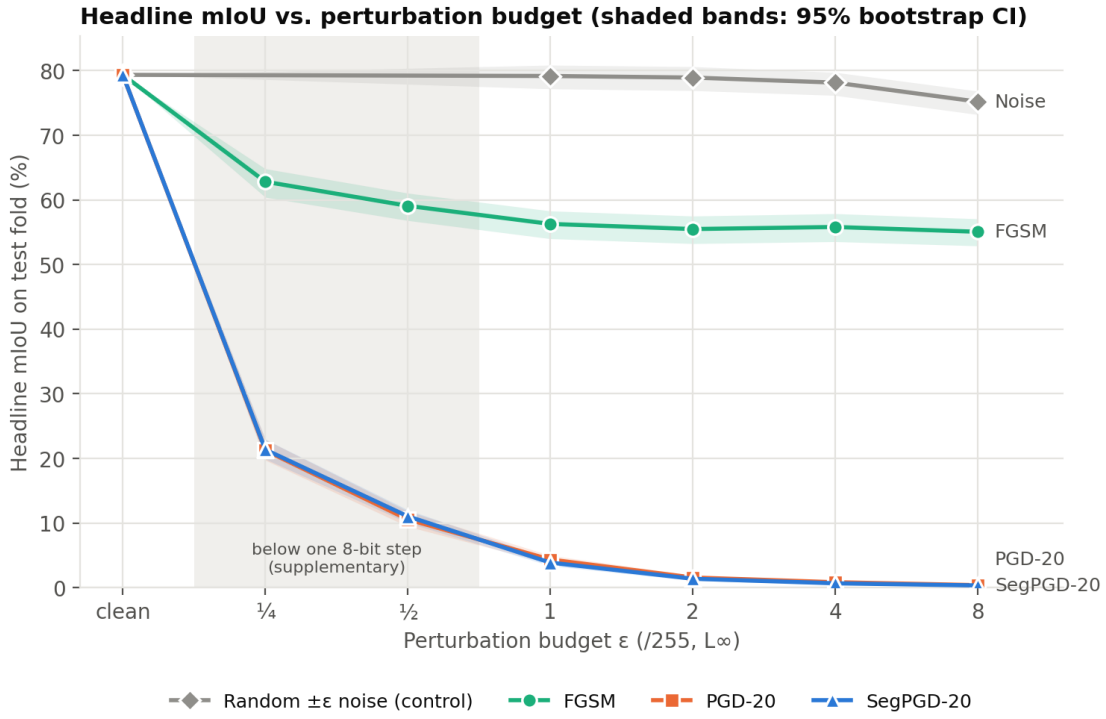


Figure 2. Headline mIoU versus perturbation budget for the three attacks and the noise control (shaded: 95% bootstrap CI). The grey band marks the supplementary budgets below one 8-bit step.

Reading the grid

- **Noise versus gradient attacks.** At ε = 1/255 the noise control and FGSM have identical PSNR (48.2 dB, every pixel moved by exactly ±1/255), yet noise costs 0.2% of mIoU and FGSM 29.1%. The vulnerability is directional, not a general sensitivity to small perturbations.
- **FGSM plateaus.** Its damage barely changes with ε (29.1% → 30.1% → 29.7% → 30.6%). A single linear step stops gaining once the loss surface curves; a larger budget does not substitute for

iterations.

- **Iterative attacks saturate at the smallest specified budget.** PGD-20 and SegPGD-20 already remove 95% of mIoU at ε = 1/255, which motivated the supplementary sub-8-bit budgets (Section 5.6).
- **PGD perturbations are less visible than FGSM's** at the same ε (51.2 vs 48.2 dB at 1/255) — PGD does not push every pixel to ±ε — yet they do about three times the damage.

5.3 Attack strength and number of iterations

Table 15. Step-count ablation at ε = 4/255 (specification); α = ε/4 for every step count.

Steps	1 (FGSM)	5	10	20	40
PGD mIoU %	55.81	9.71	2.73	0.84	0.34
SegPGD mIoU %	—	8.99	2.48	0.68	0.33
PGD correct→wrong %	2.6	21.7	45.8	69.1	85.3
SegPGD correct→wrong %	—	25.0	51.2	75.7	91.2

Table 16. Step-count ablation at ε = 1/255 (supplementary); α = ε/4 for every step count.

Steps	1 (FGSM)	5	10	20	40
PGD mIoU %	56.28	22.01	9.38	4.35	2.24
SegPGD mIoU %	—	20.97	9.61	3.88	2.57
PGD correct→wrong %	2.7	15.0	37.1	62.5	80.1
SegPGD correct→wrong %	—	17.7	42.0	68.2	86.1

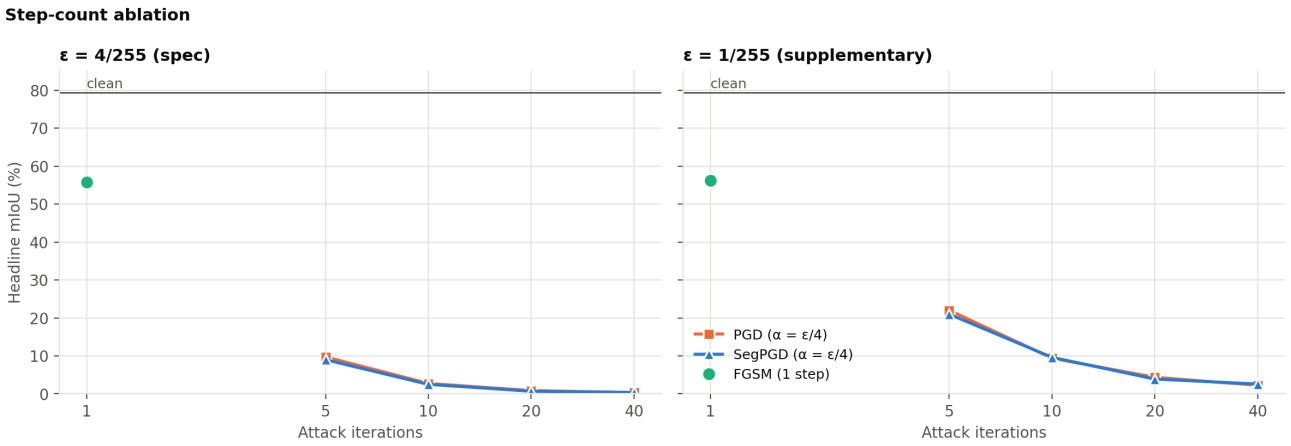


Figure 3. Headline mIoU versus number of attack iterations at ε = 4/255 (left, specification) and 1/255 (right, supplementary).

- Five PGD steps at one intensity level already do more damage (−72% relative) than FGSM at eight levels (−31%): iterations matter far more than budget.
- Damage keeps increasing up to 40 steps, so the 20-step results are conservative. Measured mIoU under attack is an upper bound on the model's robustness, never proof of it (Croce et al., SEA).
- SegPGD and PGD reach statistically indistinguishable mIoU at every step count, but SegPGD consistently converts 3–7 percentage points more of the correctly classified pixels — exactly what its reweighted loss is designed to do. On this model plain PGD is already strong enough to collapse mIoU, so the segmentation-aware reweighting shows in the pixel-flip rate rather than in mIoU.

5.4 Anatomy versus instruments

This comparison — whether delicate metallic instruments degrade faster than the surrounding biological structures within the same frame — is the project's main novelty claim (RQ3).

Table 17. Relative fall of the anatomy and instrument group mIoU, their difference (instrument − anatomy) with a paired bootstrap CI, the same difference when each class is scored only on frames where it is present (removes hallucinated pixels), and the number of test videos where instruments fall further. ¼ and ½ /255 are supplementary budgets. All budgets: Appendix B.

Attack	ε	Anatomy rel. fall %	Instrument rel. fall %	Difference pp (95% CI)	Present-frames -only diff. pp	Videos (of 6)
Noise	8/255	2.8	6.4	+3.6 (+2.2 to +5.1)	+3.9	6/6
FGSM	¼/255	15.3	23.6	+8.2 (+5.2 to +11.2)	+7.1	6/6
FGSM	1/255	21.2	33.0	+11.8 (+8.5 to +15.1)	+10.2	6/6
FGSM	8/255	19.5	36.1	+16.7 (+13.4 to +20.0)	+15.3	6/6
PGD-20	¼/255	59.4	80.1	+20.7 (+16.8 to +24.2)	+14.6	6/6
PGD-20	½/255	78.1	91.0	+12.8 (+9.8 to +15.6)	+8.8	6/6
PGD-20	1/255	88.9	97.3	+8.4 (+6.4 to +10.3)	+7.8	6/6
PGD-20	8/255	99.1	99.7	+0.7 (+0.2 to +1.5)	+0.3	4/6
SegPGD-20	¼/255	62.3	78.4	+16.1 (+12.5 to +19.6)	+12.5	6/6
SegPGD-20	½/255	80.0	89.1	+9.1 (+6.3 to +11.8)	+8.1	6/6
SegPGD-20	1/255	91.2	97.0	+5.8 (+4.2 to +7.4)	+6.0	6/6
SegPGD-20	8/255	99.2	99.8	+0.6 (+0.2 to +1.2)	+0.4	5/6

- **Consistent across attack types.** Wherever the attack is not saturated (noise at 8/255, FGSM at every budget, PGD/SegPGD up to 2/255), instruments lose a larger share of their clean IoU than anatomy: the CIs exclude zero, the per-frame sign tests give $p \leq 0.025$ ($p < 0.001$ in all but one setting), and all six test videos agree.
- **Mostly genuine segmentation failure.** Excluding hallucinated pixels shrinks the gap at ¼/255 (+20.7 → +14.6 pp for PGD-20) — hallucination contributes — but a large gap remains: the attacked model segments the *real* instruments worse than the real anatomy.
- **At saturation the distinction disappears.** At $\varepsilon \geq 4/255$ both groups are near zero and the gap is ≤ 1.6 pp.
- **Relative versus absolute.** Under strong iterative attacks anatomy loses more *absolute* IoU (≈ 86 vs ≈ 75 points) simply because it starts higher; the hypothesis concerns how fast each group degrades, so relative fall is the primary measure. Both are tabulated in Appendix B.

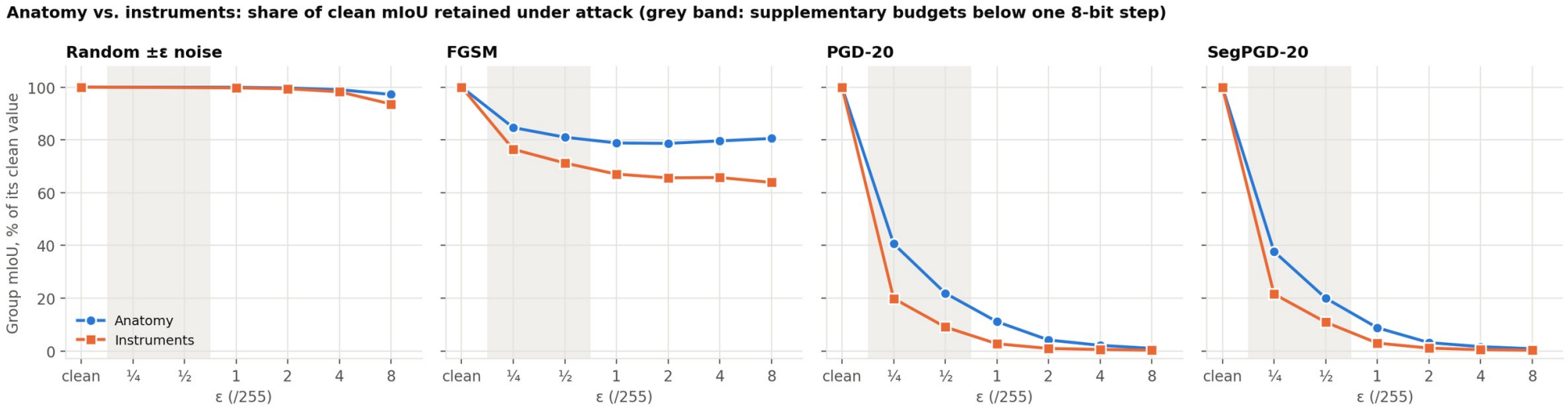


Figure 4. Share of clean group mIoU retained under each attack. The instrument line (orange) lies below the anatomy line (blue) under every attack; the gap is largest at small budgets. Grey band: supplementary budgets below one 8-bit step.

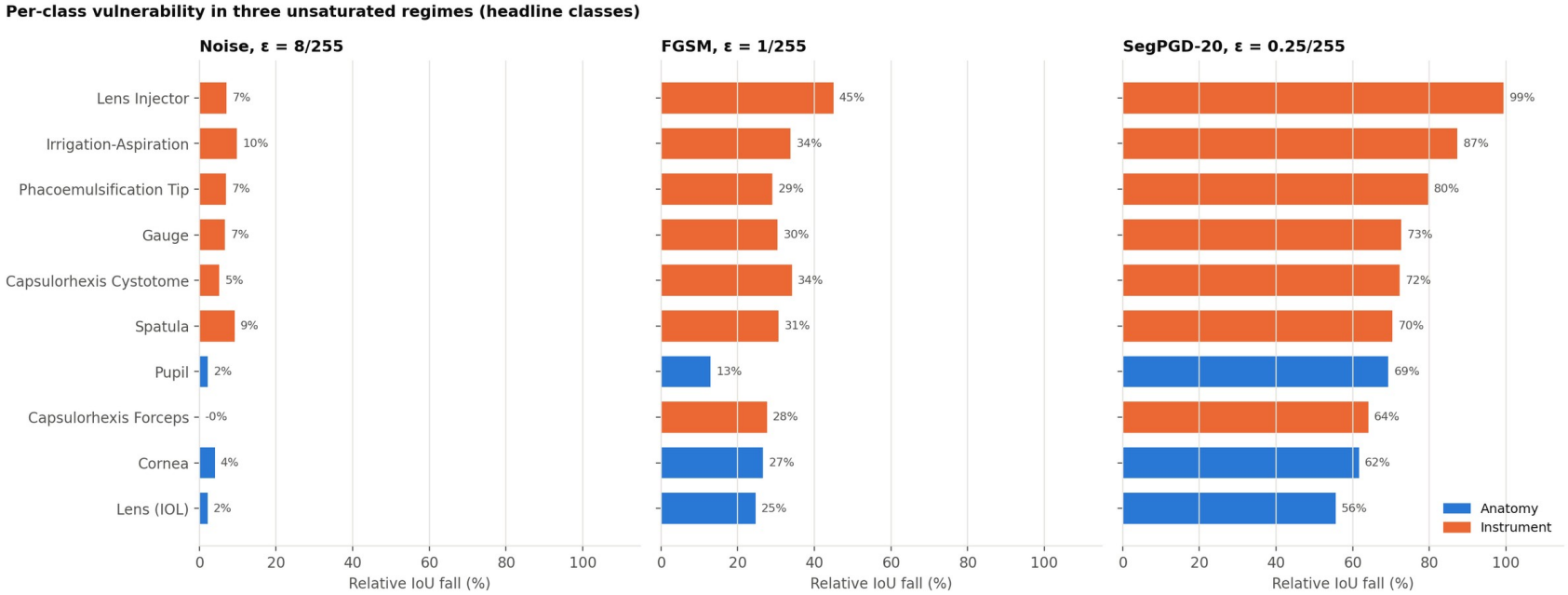


Figure 5. Per-class relative IoU fall in three unsaturated regimes (headline classes; blue = anatomy, orange = instrument). The anatomy classes are among the least affected in every regime; Capsulorhexis Forceps is the exception among instruments.

5.5 Per-class vulnerability and the “thin metallic tools” hypothesis

Table 18. Per-class relative IoU fall (%) in the unsaturated regimes. * rare class (excluded from headline means; Katena Forceps is near 0% IoU even when clean, so its relative fall is not meaningful). Absolute IoU for every class and run: Appendix A.

Class	Group	Test frames	Noise 8/255	FGSM 1/255	PGD-20 ¼/255	SegPGD-20 ¼/255	SegPGD-20 1/255
Cornea	Anatomy	561	4.1	26.6	57.2	61.7	91.2
Pupil	Anatomy	561	2.2	13.0	66.3	69.3	94.9
Lens (IOL)	Anatomy	107	2.2	24.7	54.1	55.5	87.3
Slit/Incision Knife *	Instrument	12	-1.7	21.9	82.8	81.1	91.9
Gauge	Instrument	103	6.6	30.4	72.9	72.6	97.9
Capsulorhexis Cystotome	Instrument	20	5.2	34.1	69.6	72.3	98.2
Capsulorhexis Forceps	Instrument	33	-0.4	27.6	69.1	64.1	98.7
Katena Forceps *	Instrument	8	1.6	0.9	-1.1	-1.9	18.6
Spatula	Instrument	212	9.2	30.7	76.4	70.3	92.8
Phacoemulsification Tip	Instrument	148	6.9	29.0	82.6	79.6	94.2
Irrigation-Aspiration	Instrument	120	9.8	33.8	87.3	87.3	98.0
Lens Injector	Instrument	14	7.0	45.0	99.0	99.3	99.8

- **The three anatomy classes are the least affected in every regime.** Under FGSM at 1/255 the pupil, lens and cornea lose 13–27% while instruments lose 28–45%; under SegPGD-20 at ¼/255 anatomy loses 56–69% and instruments 64–99% (Lens Injector −99%, Irrigation-Aspiration −87%).
- **One instrument behaves like anatomy.** Capsulorhexis Forceps is consistently among the most robust classes despite being thin and metallic.
- **Object size.** Across the 10 headline classes the Spearman correlation between mean object area and relative fall is negative in every non-trivial run (noise 8/255: $\rho = -0.20$, FGSM 1/255: $\rho = -0.45$, FGSM 8/255: $\rho = -0.35$, PGD ¼/255: $\rho = -0.25$, PGD 1/255: $\rho = -0.43$, SegPGD ¼/255: $\rho = -0.21$); smaller structures tend to fall further. With 10 classes none of these correlations is statistically significant (smallest $p = 0.09$; Appendix C).

Verdict on the hypothesis “thin, specular metallic tools degrade faster under small noise caps.” *Supported at the group level:* instruments degrade faster than anatomy under every attack type at small budgets, with CIs excluding zero, unanimous across videos, and not explained by hallucination. *Size is a plausible but unproven explanation:* the size–vulnerability correlation has a consistent negative sign but is not significant, and Capsulorhexis Forceps — a thin tool that stays robust — shows that thinness alone does not determine vulnerability.

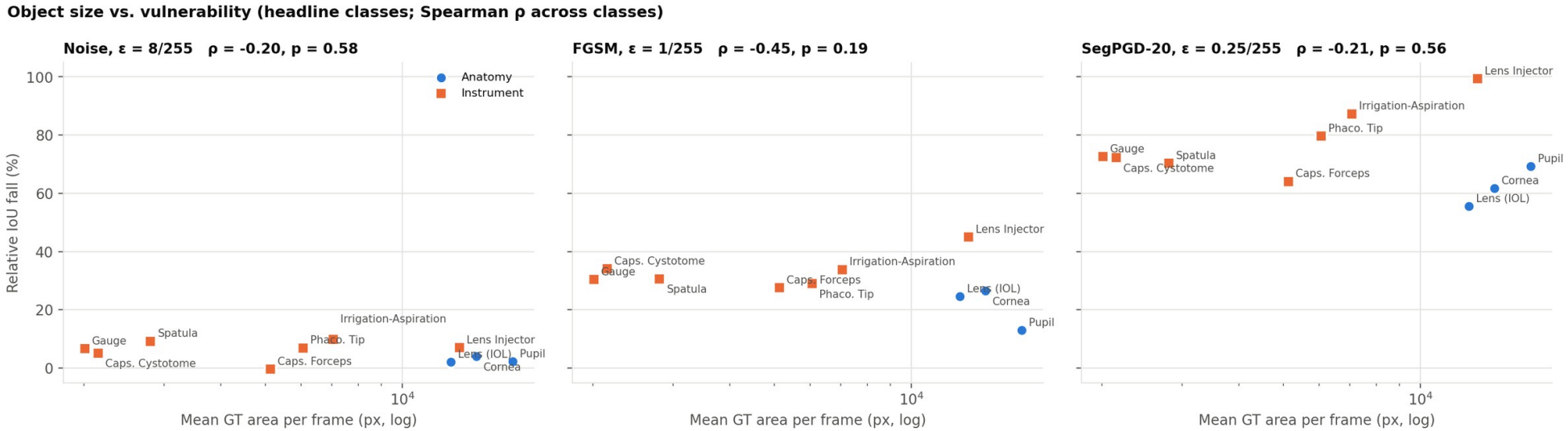


Figure 6. Mean ground-truth area per frame versus relative IoU fall, per headline class, in three unsaturated regimes (Spearman ρ across classes in the panel titles).

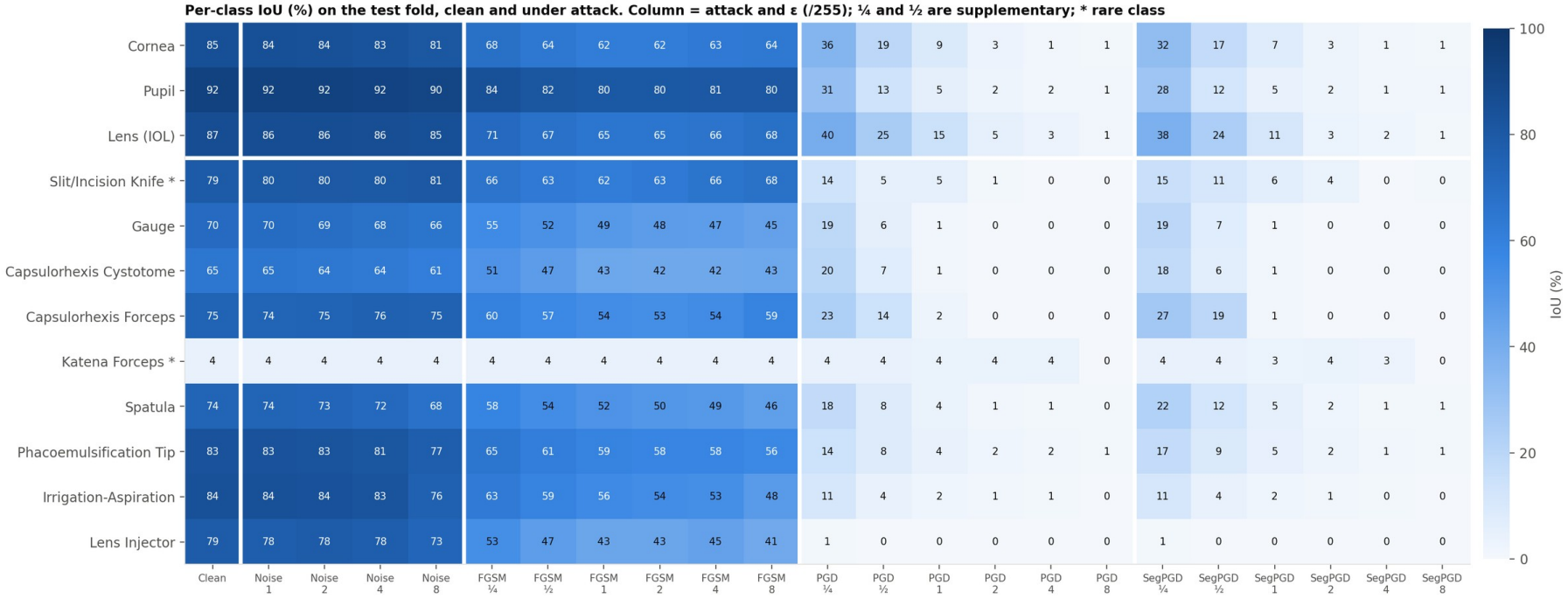


Figure 7. Per-class IoU (%) on the test fold, clean and under every attack and budget (¼ and ½ /255 are supplementary; * rare class).

5.6 Below one 8-bit step, and 8-bit quantisation

The specification's smallest budget (1/255) already saturates PGD and SegPGD. Two questions follow: where does the collapse begin, and does the attack still work once the adversarial frame is stored as an ordinary 8-bit image?

Table 19. Budgets below one 8-bit step (¼, ½ /255; supplementary) with 1/255 for reference. These perturbations exist only for an attacker who can modify decoded frames in memory.

Attack	ε	mIoU % (95% CI)	Rel. fall %	Anatomy mIoU %	Instrument mIoU %	Correct→wrong %
FGSM	¼/255	62.81 (60.41–64.81)	20.8	74.36	57.86	1.9
FGSM	½/255	59.08 (56.81–61.04)	25.5	71.15	53.91	2.4
FGSM	1/255	56.28 (54.00–58.29)	29.1	69.24	50.73	2.7
PGD-20	¼/255	21.23 (19.72–22.92)	73.2	35.67	15.04	23.9
PGD-20	½/255	10.56 (9.46–11.81)	86.7	19.21	6.85	44.9
PGD-20	1/255	4.35 (3.71–4.98)	94.5	9.73	2.04	62.5
SegPGD-20	¼/255	21.36 (19.90–22.95)	73.1	33.10	16.32	26.3
SegPGD-20	½/255	11.03 (10.03–12.03)	86.1	17.56	8.24	50.0
SegPGD-20	1/255	3.88 (3.39–4.42)	95.1	7.71	2.24	68.2

Table 20. 8-bit quantisation check: the adversarial frame is rounded to the 8-bit grid (as when saved to PNG) before inference. Clean frames are already on that grid, so the budget still holds exactly.

Attack	ε	mIoU % (float)	mIoU % (8-bit)	Change pp	max ‖δ‖∞ after rounding
FGSM	1/255	56.28	56.28	-0.00	1.00/255
FGSM	2/255	55.49	55.48	-0.00	2.00/255
PGD-20	1/255	4.35	4.09	-0.26	1.00/255
PGD-20	2/255	1.57	1.65	+0.08	2.00/255
SegPGD-20	1/255	3.88	3.75	-0.13	1.00/255
SegPGD-20	2/255	1.38	1.53	+0.14	2.00/255

- A quarter of one intensity level removes 73% of mIoU with 20 iterations; the ε curve (Figure 2) is steepest *below* the specified range, so the model has no meaningful robustness margin at any budget an 8-bit attacker can use.
- Rounding changes mIoU by at most 0.26 points: the headline result holds for adversarial frames stored as standard 8-bit images and is not a floating-point artefact.

5.7 What the model predicts under attack

Table 21. Share of all test pixels (%) assigned to each class: ground truth and predictions.

Class	Ground truth	Clean	Noise 8/255	FGSM 4/255	PGD-20 4/255	SegPGD-20 4/255
Background	84.8	84.9	85.1	85.1	40.3	33.6
Cornea	5.5	5.5	5.5	5.5	5.6	6.3
Pupil	6.7	6.6	6.7	6.8	43.7	49.5
Lens (IOL)	0.9	0.9	0.9	0.8	0.4	0.4
Slit/Incision Knife	0.0	0.0	0.0	0.0	1.3	0.6
Gauge	0.1	0.1	0.1	0.1	1.1	0.6
Capsulorhexis Cystotome	0.0	0.0	0.0	0.0	0.0	0.0
Capsulorhexis Forceps	0.1	0.1	0.1	0.1	0.0	0.0
Katena Forceps	0.0	0.0	0.0	0.0	0.0	0.0
Spatula	0.4	0.4	0.3	0.3	0.9	1.1
Phacoemulsification Tip	0.6	0.6	0.6	0.5	0.7	1.1
Irrigation-Aspiration	0.6	0.6	0.5	0.5	0.6	0.7
Lens Injector	0.1	0.1	0.1	0.1	5.4	6.0

FGSM leaves the class proportions unchanged while removing about 30% of the mIoU: it *mislocalises and confuses* structures. PGD and SegPGD make the model *hallucinate*: almost half of the frame becomes “pupil” and about 6% becomes “lens injector”, an instrument present in only 14 test frames. For any downstream use — tool tracking, phase recognition, skill assessment — confident false structures are a worse failure than an empty mask.

5.8 Qualitative examples

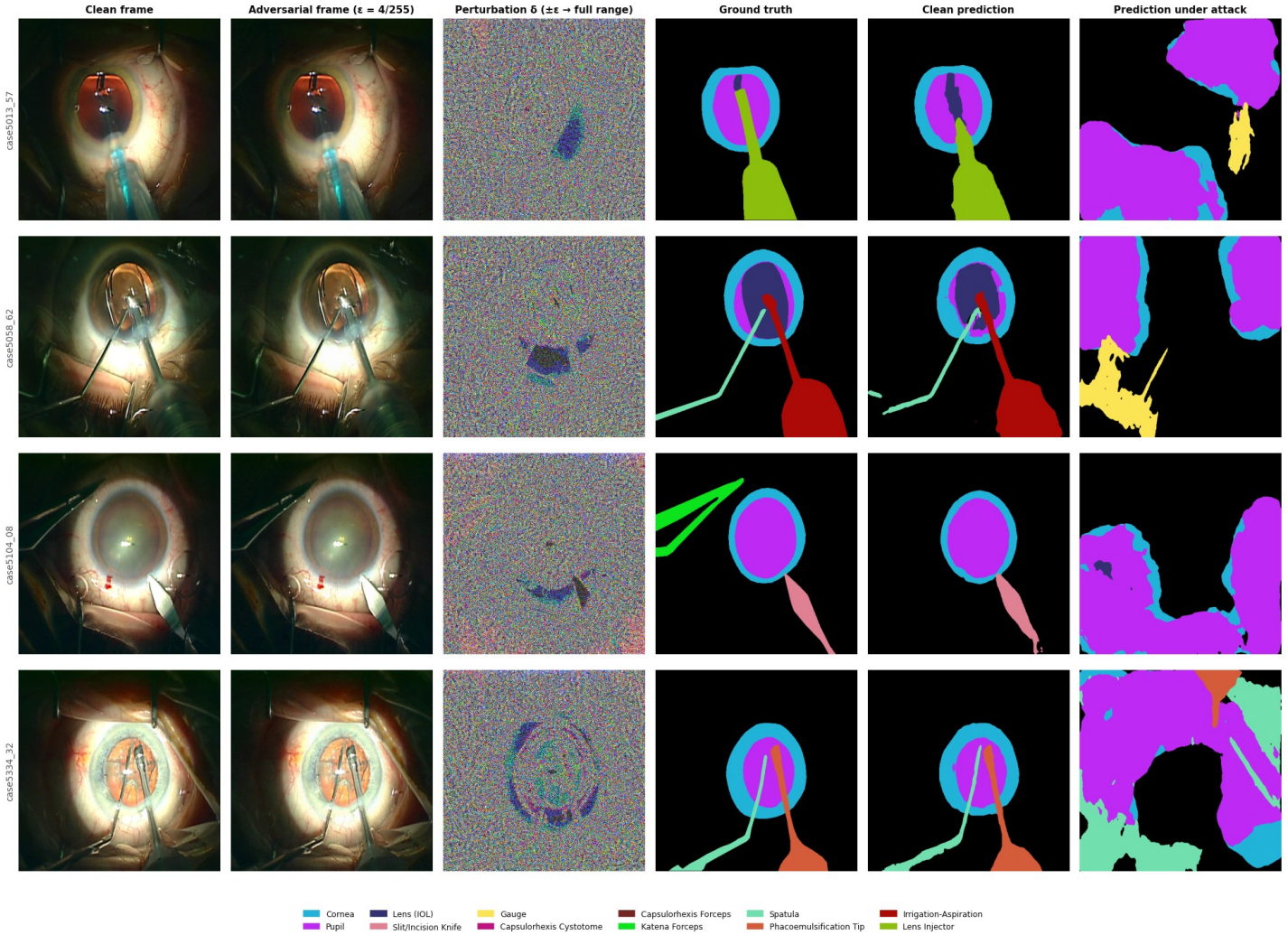


Figure 8. SegPGD-20 at ε = 4/255 on four test frames. Columns: clean frame, adversarial frame, perturbation (rescaled from ±ε to the full range so it is visible), ground truth, clean prediction, prediction under attack. The adversarial frames are visually indistinguishable from the clean ones, yet the predictions become large hallucinated regions. Row 3: the Katena forceps (green, top left of the ground truth) is missed even by the clean model.

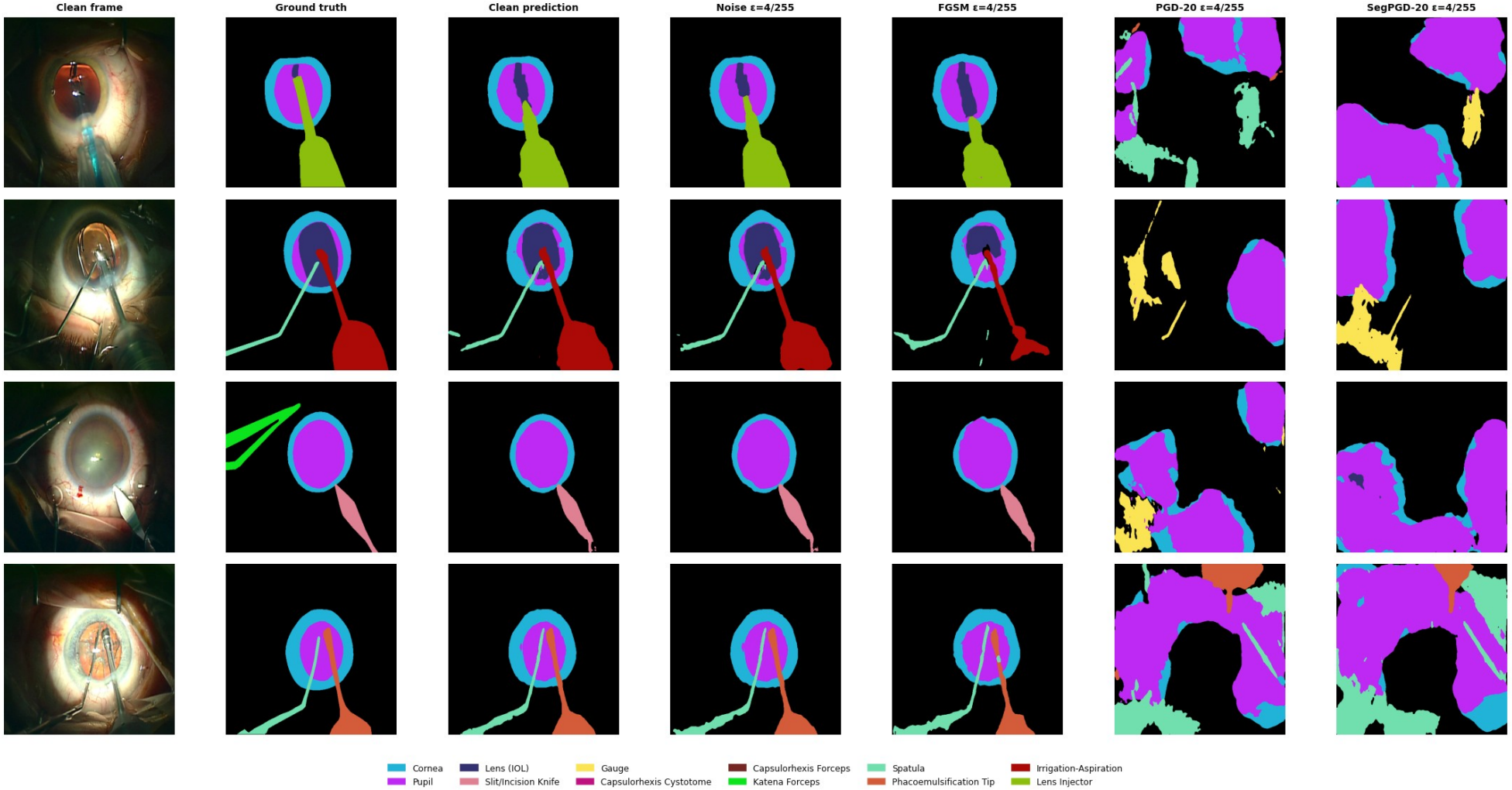


Figure 9. The same four frames under each attack at ε = 4/255: random noise and FGSM preserve most structure, PGD-20 and SegPGD-20 destroy it.

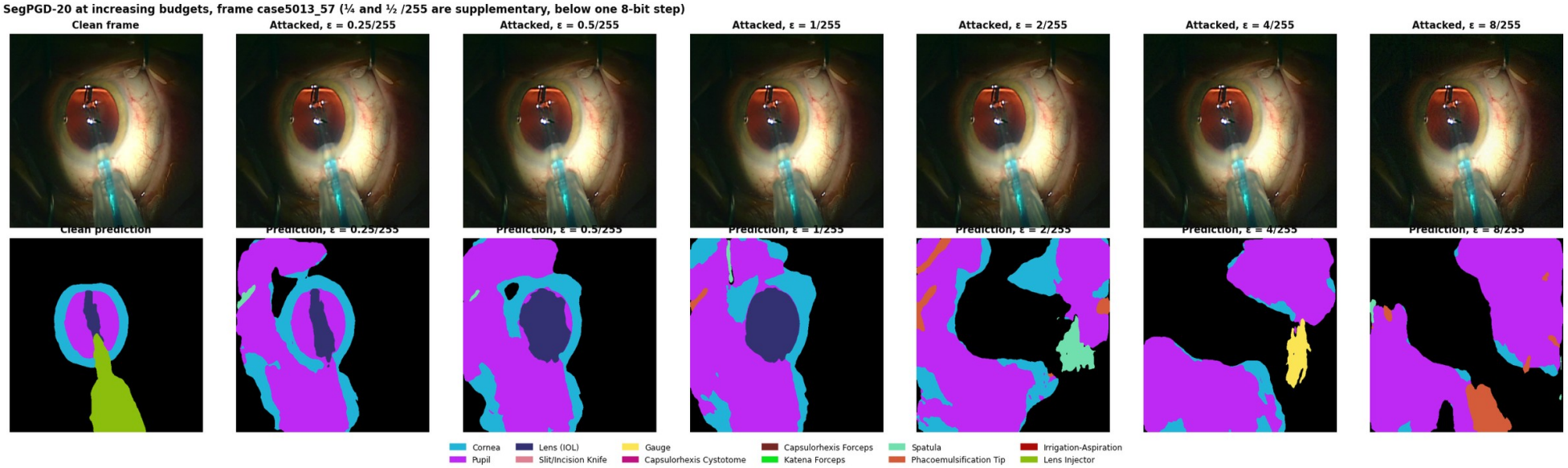


Figure 10. One frame under SegPGD-20 at increasing budgets (top: attacked frames; bottom: predictions). The input barely changes while the output collapses already at ¼/255.

5.9 Replication on the first 908-frame run

The full pipeline was first run on a partial download of 908 frames (524 train / 259 test, 80 epochs); its complete write-up is archived in `results_908frames/`. Comparing the two independent runs shows which conclusions are robust.

Table 22. Replication: the 908-frame run (2.7× less training data, 2.2× fewer test frames) versus the full-data run.

Measure	908-frame run	Full data	Same conclusion?
Clean headline mIoU	77.08	79.35	✔ competent baseline
Noise 8/255, relative fall	10.3%	5.2%	— more data halved noise sensitivity
FGSM 1/255, relative fall	34.0%	29.1%	✔ about one third, flat in ε
PGD-20 1/255, mIoU (rel. fall)	3.85 (95.0%)	4.35 (94.5%)	✔ collapse at one intensity level
SegPGD-20 1/255, mIoU (rel. fall)	3.74 (95.1%)	3.88 (95.1%)	✔
8-bit rounding, max change	0.37 pp	0.26 pp	✔ survives quantisation
Pupil share under SegPGD-20 4/255	41.2%	49.5%	✔ hallucination
Instrument − anatomy gap, FGSM 1/255	+9.7 pp, 6/6 videos, $p < 0.001$	+11.8 pp, 6/6 videos, $p < 0.001$	✔
Instrument − anatomy gap, PGD-20 ¼/255	+8.7 pp, 4/6 videos, $p = 0.55$	+20.7 pp, 6/6 videos, $p < 0.001$	✘ inconclusive → clear
Size–vulnerability ρ, PGD-20 / SegPGD-20 ¼/255	+0.40 / +0.60	-0.25 / -0.21	✘ sign flipped

Every headline vulnerability result replicates. Two secondary findings changed, and in both cases the full-data result — larger training and test sets, all classes evaluable — supersedes the 908-frame one: the anatomy/instrument asymmetry under iterative attacks, inconclusive on 908 frames, is clear on the full data; and the apparent reversal of the size effect under iterative attacks did not replicate. More training data made the model more robust to random noise but not to adversarial perturbation.

6. Answers to the Research Questions

Primary — How do adversarial attacks affect the performance and robustness of the model?
Catastrophically, and at very small budgets. A white-box iterative attack with ε = 1/255 (one intensity level, ≈ 51 dB PSNR) removes 95% of the headline mIoU (79.35% → 3.88–4.35%), and the attack survives being saved as an 8-bit image. At 8/255 the model is at 0.3–0.4%. A single-step attack removes about 30%; random noise of the same budget at most 5.2%.

Table 23. Answers to the supporting research questions.

#	Answer	Evidence
RQ1	ΔmIoU 23–24 points under FGSM and 75–79 points under PGD-20 / SegPGD-20; mean Dice falls from 88.3% to 70–71% (FGSM) and 0.7–8% (iterative).	§5.2
RQ2	Iterations matter far more than budget: SegPGD ≈ PGD ≫ FGSM ≫ noise. More steps keep helping up to 40. FGSM barely improves with ε. SegPGD flips 3–7 points more correct pixels than PGD.	§5.2, §5.3
RQ3	Yes. Instruments lose a larger share of IoU than anatomy at every unsaturated budget of every attack (+3.6 to +20.7 points; all six videos), mostly from genuine segmentation failure. Lens Injector and Irrigation-Aspiration are the most fragile classes; the anatomy classes and Capsulorhexis Forceps the most robust. Smaller structures tend to fall further (consistent sign, not significant).	§5.4, §5.5
RQ4	79.35% clean → 55.1–56.3% (FGSM) and 0.3–4.4% (iterative) headline mIoU; full per-class tables in Appendix A.	§5.2, App. A
RQ5	Strongly non-linear. Iterative attacks remove 73% of mIoU at ¼/255 and 95% at 1/255, then flatten; FGSM reaches its ∼30% plateau by 1/255; noise grows slowly (0.2% → 5.2% from 1 to 8/255). Results are unchanged after 8-bit rounding.	§5.2, §5.6

7. Verification Against the Problem Statement and Plan

7.1 Problem statement

Status: answered. The vulnerability of a cataract surgical segmentation model to adversarial perturbation has been quantified on the complete official held-out fold, with confidence intervals, a no-gradient control, an 8-bit storage check and an independent replication (Sections 5 and 6).

7.2 Specific objectives (specification §7)

Table 24. Objectives of the Version 1 specification.

#	Objective	Status	Evidence
1	Study existing research on cataract surgical AI and adversarial attacks	Done (before this phase)	Literature pool, gap matrix
2	Select an appropriate dataset and AI model	Done	Cataract-1K (2,256 frames); DeepLabV3+ / ResNet-50
3	Establish baseline performance on clean inputs	Done	79.35% mIoU, 88.27% mDice (§5.1)
4	Implement adversarial attack methods	Done	FGSM, PGD, SegPGD + noise control (§4.5)
5	Generate adversarial examples	Done	40 attacked runs × 561 frames
6	Evaluate performance on adversarial inputs	Done	Same model and metrics, fp32 (§5.2)
7	Quantify degradation relative to the clean baseline	Done	ΔmIoU, relative fall, robustness ratio, paired CIs
8	Analyse attack strength versus degradation	Done	6 budgets, two step ablations (§5.2–5.3, §5.6)
9	Identify vulnerability patterns	Done	Anatomy vs instruments, per class, size, hallucination
10	Document findings and limitations	Done	This document

7.3 Locked decisions (specification §25) and plan milestones

Table 25. Compliance with the locked Version 1 decisions and the implementation plan.

Requirement	Status
DeepLabV3+, ImageNet ResNet-50, OS16, 13 channels, 512×512	✔ (earlier code used DeepLabV3; corrected)
Official patient-disjoint folds, Fold 1 test, 20 train / 4 validation videos	✔ (earlier code tested on Fold 2; corrected)
White-box, untargeted, L∞, perturbation in [0,1] RGB before normalisation, clipped	✔ checked on every batch
FGSM; PGD-20 (α = ε/4, random start); SegPGD-20 as the primary attack	✔
ε ∈ {1, 2, 4, 8}/255 → 12 attacked runs + 1 clean	✔ (+ 4 noise controls)
Step-count ablation {5, 10, 20, 40} at ε = 4/255 for PGD and SegPGD	✔ (+ the same at 1/255)
mIoU and mean Dice headline; per-class IoU always reported; rare classes separate	✔
Absolute and relative degradation, robustness ratio, mIoU-vs-ε curves	✔
Anatomy versus instrument comparison	✔ with CIs, per-video consistency, hallucination control
Clean baseline within the published benchmark range (plan, Phase 2)	✔ meets or exceeds every value
40 training epochs (plan, Phase 2)	✔
Unit-tested perturbation clipping (plan, Phase 3)	✔ self-test + runtime assertion
Figure panels: clean, perturbed, difference map, clean and attacked prediction (plan, Phase 4)	✔ §5.8
Documentation, requirements, fixed seeds, README (plan, Phase 5)	✔
No defences or adversarial training in Version 1 (specification §28)	✔ none added
Thesis-format final report and presentation slides (plan, Phases 5–6)	Open — this document supplies the material

8. Limitations and Threats to Validity

- **One fold, one seed, one architecture.** Fold 1 only, as the specification requires; no cross-fold variance and no second architecture (the optional UNet++ arm was not run). The 908-frame replication (§5.9) partly offsets this.
- **Correlated frames.** Bootstrap CIs and per-frame sign tests treat frames as independent; frames from the same video are not. The six-video consistency counts are the conservative evidence.
- **Rare classes.** Slit/Incision Knife (12 test frames) and Katena Forceps (8) are too sparse for firm per-class conclusions; Katena is barely segmented even on clean frames (4.1% IoU).
- **Attack strength is a lower bound.** Damage was still rising from 20 to 40 iterations; measured mIoU under attack is an upper bound on robustness, never proof of it.
- **Threat model.** White-box, digital, untargeted, with ground-truth labels available to the attacker — the worst case by design, not a realistic operating-room attack. Black-box, transfer, physical and temporal attacks are Version 2.
- **Sub-8-bit budgets** (¼, ½ /255) cannot be stored in an 8-bit image and apply only to in-memory manipulation of decoded frames; the 8-bit-rounded runs cover the stored-image case.
- **Benchmark comparison** with the Cataract-1K paper is not like-for-like (single fold versus 5-fold mean, different training recipe).
- **Timing figures are wall-clock** and were affected by GPU sharing with other applications and, for one run, by a system sleep (marked ‡ in Appendix D). No result depends on timing.
- **No clinical claims.** These are laboratory measurements on recorded frames; they say nothing directly about patient safety.

9. Deviations from the Plan

Table 26. Deviations from the original plan. All additions are measurements; none is a defence, so the Version 1 scope is preserved.

Deviation	Reason
Random ±ε noise control added	Separates “any small perturbation” from “adversarial direction”; without it the degradation is uninterpretable
Sub-8-bit budgets (¼, ½ /255) and 8-bit-rounded runs added	Iterative attacks saturated at the smallest specified budget, leaving no unsaturated iterative points
Step ablation also at 1/255	To see the effect of iterations before saturation
Prediction-composition table and present-frames-only control	Explain how the model fails (hallucination versus mislocalisation) and test whether the instrument gap is real
First run on a 908-frame partial download, kept as replication	The first download was truncated by Synapse’s 1 GB package limit; the study was repeated on all 2,256 frames

10. Reproducibility

10.1 Environment

Table 27. Software and hardware used for all reported results.

Component	Version / specification
Hardware	NVIDIA GeForce RTX 3060 Laptop GPU (6 GB), 16 GB RAM, Windows 11
Python	3.12
Deep learning	PyTorch 2.5.1 + CUDA 12.1, torchvision 0.20.1
Scientific stack	NumPy 1.26.2, SciPy 1.13.1, pandas 2.1.3, matplotlib 3.8.2, Pillow 10.1.0
Data download	synapseclient 4.14.0 (separate virtual environment)
This document	ReportLab 5.0.1, DejaVu fonts

10.2 Commands

```
python download_synapse_dataset.py # fetch frames (token in ~/.synapseConfig)

python convert_supervisely_to_masks.py # build the 13-class masks

python cataract_dataset.py # verify folds, split and class counts

python train_baseline.py --epochs 40 # train the clean baseline

python run_experiments.py # 41-run grid (resumable)

python analyze_results.py # tables and figures

python report/build_results_pdf.py # this document
```

Self-tests: `python loss.py`, `python metrics.py`, `python model_builder.py`, `python attacks.py`.

10.3 Artefacts

Table 28. Where the outputs live (project root).

Path	Contents
runs/baseline/	best_model.pth (epoch 36), history.csv, training_curves.png, clean_report.txt
results/raw/	Per-run, per-frame, per-class TP/FP/FN counts and attack statistics (41 .npz files) — the source of every number
results/tables/, results/figures/	Generated tables (CSV) and figures (PNG)
results/RESULTS_TABLES.md, results/summary.json	Auto-generated tables and key numbers
results/qualitative/	Perturbations and predictions for the four qualitative frames
results_908frames/, runs/baseline_908frames/	The replication run on 908 frames
Results_and_Verification.md	Markdown version of the findings
report/	This PDF and its build script

10.4 Run times

Training took 54.7 minutes. The 41 evaluation runs took 5.1 hours of attack compute (excluding one run interrupted by a system sleep), with the GPU partly shared with other applications. Per-run times are listed in Appendix D.

Appendix A. Per-class IoU for Every Run

Table 29. IoU (%) per class (columns) for every clean and attacked run (rows). * rare class. PGD/SegPGD: 20 steps. Step-ablation and quantised runs: Appendix D.

Run	ε	Cornea	Pupil	Lens	Knife*	Gauge	Cyst.	Caps. F.	Katena*	Spatula	Phaco	Irr.-Asp.	Injector	Headline mIoU
Clean	—	84.5	92.4	86.6	79.2	70.4	64.6	75.0	4.1	74.4	82.9	84.1	78.6	79.35
Noise	1/255	84.4	92.4	86.5	79.7	70.0	65.1	74.3	4.1	74.4	82.8	84.1	77.6	79.17
Noise	2/255	84.1	92.2	86.2	79.9	69.3	64.4	74.9	4.1	73.4	82.7	83.6	78.3	78.93
Noise	4/255	83.3	91.8	85.8	79.6	67.9	63.7	75.9	4.0	71.6	81.4	82.5	77.5	78.15
Noise	8/255	81.1	90.4	84.7	80.5	65.7	61.3	75.3	4.0	67.5	77.2	75.8	73.1	75.20
FGSM	¼/255	68.2	84.1	70.7	66.2	55.4	51.3	59.6	4.1	57.9	64.9	62.7	53.4	62.81
FGSM	½/255	64.2	81.9	67.3	63.2	52.0	47.4	56.5	4.1	54.2	61.2	58.9	47.1	59.08
FGSM	1/255	62.1	80.4	65.2	61.8	49.0	42.6	54.2	4.0	51.5	58.8	55.7	43.2	56.28
FGSM	2/255	61.8	80.2	65.2	63.0	47.6	42.1	53.2	4.0	50.0	57.7	54.0	43.0	55.49
FGSM	4/255	62.9	80.6	66.2	65.6	46.9	42.1	54.5	4.0	48.8	57.7	53.3	45.0	55.81
FGSM	8/255	63.7	80.5	68.0	67.7	44.9	42.8	58.8	3.9	46.3	56.0	48.2	41.4	55.06
PGD-20	¼/255	36.2	31.2	39.7	13.6	19.1	19.7	23.2	4.1	17.5	14.4	10.7	0.8	21.23
PGD-20	½/255	19.3	13.3	25.1	5.2	5.8	7.1	14.4	4.0	8.3	8.0	4.2	0.2	10.56
PGD-20	1/255	9.0	5.5	14.7	5.3	1.5	1.0	2.1	3.8	3.7	4.3	1.6	0.1	4.35
PGD-20	2/255	3.1	2.4	5.4	0.6	0.1	0.3	0.0	3.8	1.0	1.9	1.3	0.2	1.57
PGD-20	4/255	1.2	1.5	2.7	0.0	0.0	0.0	0.0	3.7	0.5	1.5	0.6	0.3	0.84
PGD-20	8/255	0.7	0.9	0.9	0.0	0.0	0.0	0.0	0.0	0.4	0.6	0.1	0.3	0.39
SegPGD-20	¼/255	32.4	28.4	38.5	15.0	19.3	17.9	26.9	4.2	22.1	16.9	10.7	0.5	21.36
SegPGD-20	½/255	17.1	11.6	24.0	10.9	7.0	5.9	19.4	4.0	11.5	9.3	4.4	0.2	11.03
SegPGD-20	1/255	7.4	4.7	11.0	6.4	1.5	1.2	1.0	3.3	5.4	4.8	1.7	0.2	3.88
SegPGD-20	2/255	2.7	2.2	3.4	3.6	0.1	0.0	0.0	3.9	1.7	2.4	1.2	0.1	1.38
SegPGD-20	4/255	1.2	1.3	1.8	0.3	0.0	0.0	0.0	3.1	0.8	1.1	0.5	0.2	0.68
SegPGD-20	8/255	0.6	0.8	0.7	0.5	0.0	0.0	0.0	0.0	0.5	0.6	0.1	0.1	0.34

Appendix B. Anatomy versus Instruments, All Budgets

Table 30. Group-level comparison for every attack and budget. Relative fall = (clean − attacked) / clean of the group mIoU; Δ = absolute drop in points. Frames: per-frame sign test on robustness ratios (frames containing a headline instrument; one-sided). Videos: test videos in which instruments fall further. Present-only: each class scored only on frames where it occurs.

Attack	ε	Anat. rel. fall %	Instr. rel. fall %	Diff. pp (95% CI)	Anat. Δ pp	Instr. Δ pp	Frames instr. worse	Sign-test p	Videos	Present-only diff. pp
Noise	1/255	0.1	0.3	+0.2 (-0.3 to +0.7)	0.1	0.2	243/438	0.012	4/6	+0.3
Noise	2/255	0.4	0.6	+0.3 (-0.2 to +0.8)	0.3	0.5	248/438	0.003	4/6	+0.4
Noise	4/255	1.0	1.8	+0.8 (-0.2 to +1.7)	0.9	1.3	250/438	0.002	5/6	+0.9
Noise	8/255	2.8	6.4	+3.6 (+2.2 to +5.1)	2.5	4.9	281/438	<0.001	6/6	+3.9
FGSM	¼/255	15.3	23.6	+8.2 (+5.2 to +11.2)	13.5	17.8	240/438	0.025	6/6	+7.1
FGSM	½/255	19.0	28.8	+9.8 (+6.5 to +12.9)	16.7	21.8	256/438	<0.001	6/6	+8.4
FGSM	1/255	21.2	33.0	+11.8 (+8.5 to +15.1)	18.6	25.0	278/438	<0.001	6/6	+10.2
FGSM	2/255	21.3	34.4	+13.1 (+9.7 to +16.5)	18.7	26.1	295/438	<0.001	6/6	+11.3
FGSM	4/255	20.4	34.3	+13.9 (+10.3 to +17.3)	17.9	25.9	313/438	<0.001	6/6	+12.2
FGSM	8/255	19.5	36.1	+16.7 (+13.4 to +20.0)	17.1	27.4	324/438	<0.001	6/6	+15.3
PGD-20	¼/255	59.4	80.1	+20.7 (+16.8 to +24.2)	52.2	60.7	268/438	<0.001	6/6	+14.6
PGD-20	½/255	78.1	91.0	+12.8 (+9.8 to +15.6)	68.6	68.9	299/431	<0.001	6/6	+8.8
PGD-20	1/255	88.9	97.3	+8.4 (+6.4 to +10.3)	78.1	73.7	318/397	<0.001	6/6	+7.8
PGD-20	2/255	95.9	99.1	+3.2 (+2.1 to +4.4)	84.2	75.0	337/377	<0.001	6/6	+3.3
PGD-20	4/255	97.9	99.5	+1.6 (+0.8 to +2.5)	86.0	75.3	318/349	<0.001	5/6	+1.4
PGD-20	8/255	99.1	99.7	+0.7 (+0.2 to +1.5)	87.0	75.5	281/297	<0.001	4/6	+0.3
SegPGD-20	¼/255	62.3	78.4	+16.1 (+12.5 to +19.6)	54.7	59.4	257/438	<0.001	6/6	+12.5
SegPGD-20	½/255	80.0	89.1	+9.1 (+6.3 to +11.8)	70.3	67.5	273/432	<0.001	6/6	+8.1
SegPGD-20	1/255	91.2	97.0	+5.8 (+4.2 to +7.4)	80.1	73.5	324/417	<0.001	6/6	+6.0
SegPGD-20	2/255	96.9	99.0	+2.1 (+1.2 to +3.1)	85.1	74.9	347/401	<0.001	6/6	+2.1
SegPGD-20	4/255	98.4	99.5	+1.1 (+0.5 to +2.0)	86.4	75.3	349/389	<0.001	4/6	+1.0
SegPGD-20	8/255	99.2	99.8	+0.6 (+0.2 to +1.2)	87.1	75.5	303/331	<0.001	5/6	+0.4

Appendix C. Object Size versus Vulnerability

Table 31. Spearman correlation across the 10 headline classes between mean ground-truth area per frame and relative IoU fall (negative ρ: smaller structures fall further). Runs where every class is near 100% fall carry no size information.

Run	ε	Spearman ρ	p	Classes	Mean rel. fall %
Noise	1/255	+0.19	0.603	10	0.2
Noise	2/255	-0.43	0.214	10	0.6
Noise	4/255	-0.56	0.090	10	1.6
Noise	8/255	-0.20	0.580	10	5.3
FGSM	¼/255	-0.32	0.365	10	21.1
FGSM	½/255	-0.35	0.328	10	25.8
FGSM	1/255	-0.45	0.187	10	29.5
FGSM	2/255	-0.41	0.244	10	30.5
FGSM	4/255	-0.41	0.244	10	30.2
FGSM	8/255	-0.35	0.328	10	31.1
PGD-20	¼/255	-0.25	0.489	10	73.5
PGD-20	½/255	-0.26	0.467	10	86.9
PGD-20	1/255	-0.43	0.214	10	94.8
PGD-20	2/255	-0.64	0.048	10	98.1
PGD-20	4/255	-0.56	0.090	10	99.0
PGD-20	8/255	-0.70	0.025	10	99.5
SegPGD-20	¼/255	-0.21	0.556	10	73.2
SegPGD-20	½/255	-0.14	0.701	10	86.2
SegPGD-20	1/255	-0.22	0.533	10	95.3
SegPGD-20	2/255	-0.62	0.054	10	98.4
SegPGD-20	4/255	-0.67	0.033	10	99.2
SegPGD-20	8/255	-0.73	0.016	10	99.6

Table 32. Mean ground-truth area per frame (frames where the class is present).

Class	Group	Mean GT area per frame (px at 512×512)
Gauge	Instrument	2,009
Capsulorhexis Cystotome	Instrument	2,152
Spatula	Instrument	2,801
Slit/Incision Knife	Instrument	3,625
Capsulorhexis Forceps	Instrument	5,136
Phacoemulsification Tip	Instrument	6,062
Katena Forceps	Instrument	6,727
Irrigation-Aspiration	Instrument	7,063
Lens (IOL)	Anatomy	12,777
Lens Injector	Instrument	13,350
Cornea	Anatomy	14,524
Pupil	Anatomy	17,476

Appendix D. Complete Run List

Table 33. All 41 runs in execution order. ‡ wall-clock time inflated by a system sleep during the run; the result is unaffected. The measured maximum perturbation equals ε in every attacked run.

#	Run id	Attack	ε	Steps	8-bit	Frames	max ‖δ‖∞ (/255)	Headline mIoU %	Time (s)
1	clean	clean	—	0		561	0.000	79.35	14
2	noise_eps1	noise	1/255	0		561	1.000	79.17	13
3	noise_eps2	noise	2/255	0		561	2.000	78.93	13
4	noise_eps4	noise	4/255	0		561	4.000	78.15	15
5	noise_eps8	noise	8/255	0		561	8.000	75.20	14
6	fgsm_eps1	fgsm	1/255	1		561	1.000	56.28	45
7	fgsm_eps2	fgsm	2/255	1		561	2.000	55.49	47
8	fgsm_eps4	fgsm	4/255	1		561	4.000	55.81	45
9	fgsm_eps8	fgsm	8/255	1		561	8.000	55.06	45
10	pgd_eps1_T20	pgd	1/255	20		561	1.000	4.35	603
11	pgd_eps2_T20	pgd	2/255	20		561	2.000	1.57	653
12	pgd_eps4_T20	pgd	4/255	20		561	4.000	0.84	703
13	pgd_eps8_T20	pgd	8/255	20		561	8.000	0.39	639
14	segpgd_eps1_T20	segpgd	1/255	20		561	1.000	3.88	962
15	segpgd_eps2_T20	segpgd	2/255	20		561	2.000	1.38	877
16	segpgd_eps4_T20	segpgd	4/255	20		561	4.000	0.68	880
17	segpgd_eps8_T20	segpgd	8/255	20		561	8.000	0.34	765
18	pgd_eps4_T5	pgd	4/255	5		561	4.000	9.71	229
19	pgd_eps4_T10	pgd	4/255	10		561	4.000	2.73	504
20	pgd_eps4_T40	pgd	4/255	40		561	4.000	0.34	1,551
21	segpgd_eps4_T5	segpgd	4/255	5		561	4.000	8.99	199
22	segpgd_eps4_T10	segpgd	4/255	10		561	4.000	2.48	54,628 ‡
23	segpgd_eps4_T40	segpgd	4/255	40		561	4.000	0.33	1,100
24	pgd_eps1_T5	pgd	1/255	5		561	1.000	22.01	144
25	pgd_eps1_T10	pgd	1/255	10		561	1.000	9.38	287
26	pgd_eps1_T40	pgd	1/255	40		561	1.000	2.24	1,187
27	segpgd_eps1_T5	segpgd	1/255	5		561	1.000	20.97	165
28	segpgd_eps1_T10	segpgd	1/255	10		561	1.000	9.61	296

#	Run id	Attack	ε	Steps	8-bit	Frames	max ‖δ‖∞ (/255)	Headline mIoU %	Time (s)
29	segpgd_eps1_T40	segpgd	1/255	40		561	1.000	2.57	1,099
30	fgsm_eps0.25	fgsm	¼/255	1		561	0.250	62.81	39
31	pgd_eps0.25_T20	pgd	¼/255	20		561	0.250	21.23	577
32	segpgd_eps0.25_T20	segpgd	¼/255	20		561	0.250	21.36	618
33	fgsm_eps0.5	fgsm	½/255	1		561	0.500	59.08	42
34	pgd_eps0.5_T20	pgd	½/255	20		561	0.500	10.56	576
35	segpgd_eps0.5_T20	segpgd	½/255	20		561	0.500	11.03	625
36	fgsm_eps1_q8	fgsm	1/255	1	yes	561	1.000	56.28	40
37	pgd_eps1_T20_q8	pgd	1/255	20	yes	561	1.000	4.09	567
38	segpgd_eps1_T20_q8	segpgd	1/255	20	yes	561	1.000	3.75	598
39	fgsm_eps2_q8	fgsm	2/255	1	yes	561	2.000	55.48	44
40	pgd_eps2_T20_q8	pgd	2/255	20	yes	561	2.000	1.65	687
41	segpgd_eps2_T20_q8	segpgd	2/255	20	yes	561	2.000	1.53	780

Appendix E. Glossary

Table 34. Terms used in this document.

Term	Meaning
Adversarial example	An input with a small, deliberately computed perturbation that changes a model’s output
White-box	The attacker has full access to the model, its weights and gradients
Untargeted	The attacker wants any wrong output, not a specific one
L∞ budget ε	Maximum change of any single pixel value; ε = 1/255 is one 8-bit intensity level
FGSM	Fast Gradient Sign Method: one step of size ε along the sign of the loss gradient
PGD-T	Projected Gradient Descent: T small steps, each projected back into the ε-ball
SegPGD	Segmentation-specific PGD that reweights correctly and wrongly classified pixels over the steps
IoU / mIoU	Intersection over Union per class; its mean over a set of classes
Dice	2·TP / (2·TP + FP + FN); closely related to IoU
Relative fall	(clean − attacked) / clean × 100
Robustness ratio	attacked / clean
Correct→wrong	Share of pixels correct on the clean frame that the attack made wrong
PSNR	Peak signal-to-noise ratio between clean and adversarial frame; higher = less visible
Bootstrap CI	Confidence interval from resampling the test frames with replacement
Saturation	An attack so strong that the metric is already near its floor, hiding further differences
Hallucination	The model predicting a structure where none exists
OS16	Output stride 16: the encoder’s deepest features are 1/16 of the input resolution

Appendix F. Key References

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The full literature record (~65 papers) and the 22-paper gap matrix are in `Literature_Review_Paper_Pool.md` and `Paper_Gap_Matrix.md`.